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Inventor(s)

DWYER; Daniel P. et al.

SELF-SEALING RESPIRATORY FILTER AND CONDENSATE MANAGEMENT APPARATUS

Abstract

A respiratory filter and condensate management apparatus is provided for use in a breathing circuit during patient respiration. The apparatus includes a filter housing having an air inlet port and an air outlet port. A filter member is provided within the filter housing and located in an expiratory air flow path. A collection jar is removably attached to the filter housing and has a liquid reservoir to collect liquid formed by condensation in the flow of expiratory air within the filter housing when the collection jar is attached to the filter housing. A valve assembly moves to an open position when the filter housing is attached to the collection jar to allow drainage of the liquid from the filter housing. The valve assembly moves to a closed position when the filter housing is detached from the collection jar to prevent drainage of the liquid from the filter housing.

Inventors: DWYER; Daniel P. (Cary, NC), JACKSON; Christopher S. (Raleigh, NC)

Applicant: TELEFLEX MEDICAL INCORPORATED (Morrisville, NC)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/407,649 filed Aug. 20, 2021, which claims the benefit of priority to U.S. Provisional Patent Application No. 63/069,337 filed Aug. 24, 2020, the entire contents of which are incorporated herein by reference.

FIELD OF THE INVENTION

[0002] The present disclosure generally relates to respiratory filters for use in a breathing circuit, and more particularly, to respiratory filters operable to manage an accumulation of condensate produced during active humidification.

BACKGROUND

[0003] Ventilators are commonly used to aid or replace a patient's respiratory function. Such ventilators are typically connected to a breathing circuit having an inspiratory limb for delivering a flow of air from the ventilator to the patient for inhalation, and an expiratory limb for returning a flow of exhaled air from the patient back to the ventilator. The flow of air is often heated and humidified to increase patient comfort and compliance during respiration.

[0004] A respiratory filter may be used in the breathing circuit between the expiratory limb and the ventilator in order to prevent the ventilator from being contaminated by bacteria, viruses, and/or other waste material exhaled by the patient during respiration. The exhaled air from the patient contains vapor that may form condensate within the expiratory limb. Similarly, vapor that reaches the respiratory filter may form condensate within the filter. Some conventional expiratory limbs may be heated, such as by a heating wire, to reduce the amount of condensate within the expiratory limb. However, conventional respiratory filters are unheated, and are therefore exposed to room air that is often colder than the air inside the filter. Thus, unwanted condensation or rainout commonly occurs within the filter when the heated and humidified air enters the colder filter.

[0005] The accumulation of condensate within the respiratory filter can saturate the filter material causing it to perform less effectively. For instance, resistance to air flow in the filter material may increase as the filter material becomes more saturated with water. Additionally, condensate that forms in the filter may be contaminated with bacteria, viruses, and/or other waste material from the patient.

[0006] Some typical respiratory filters may include a container to collect moisture on the circuit side of the filter. However, such filters routinely fill up with water and require dismantling or breaking of the breathing circuit in order to drain the excess fluid from the filter. It is advantageous to avoid taking apart the breathing circuit during respiration in order to maintain a positive end expiratory pressure (PEEP) and minimize the risk of infection. Moreover, ventilation patterns can be affected when conventional respiratory filters fill up with liquid from condensation, which can cause false triggering on the ventilator. Furthermore, known heated filters either require a ventilator that has a heating capability, a standalone filter heater, or a powered humidifier, each of which blocks viewing of the filter and are also cost prohibitive.

[0007] Accordingly, there is a clear need for a respiratory filter with a releasably attachable liquid drainage jar having self-sealing capabilities for condensate management during active humidification. In particular, the respiratory filter and condensate management apparatus of the

present disclosure mitigates the buildup of liquid within the expiratory filter, minimizes filter change-outs, helps maintain PEEP in the breathing circuit, and mitigates the risk of infection due to breaking of the circuit, among other advantages.

SUMMARY

[0008] The foregoing needs are met, to a great extent, by the present disclosure of a respiratory filter and condensate management apparatus discussed herein. The respiratory filter and condensate management apparatus may comprise a filter housing including an air inlet port, an air outlet port, a base, and a base drain, the air inlet port configured to receive a flow of expiratory air, and the air outlet port configured to output the flow of expiratory air; a filter member provided within the filter housing and located in a flow path of the expiratory air between the air inlet port and the air outlet port; a collection jar removably attached to the filter housing, the collection jar including a liquid reservoir, a spill cover, and a cover drain, the cover drain and the base drain forming a liquid passageway operable to provide fluid communication between the filter housing and the liquid reservoir when the collection jar is attached to the filter housing, and the liquid reservoir operable to collect liquid formed by condensation in the flow of expiratory air within the filter housing when the collection jar is attached to the filter housing; and a valve assembly configured to permit passage of the liquid formed by condensation in the flow of expiratory gas through the liquid passageway when the valve assembly is in an open position, and configured to prevent passage of the liquid through the liquid passageway when the valve assembly is in a closed position.

[0009] According to another aspect of the disclosure, the valve assembly is operable to automatically assume the open position when the filter housing is attached to the collection jar.

[0010] According to another aspect of the disclosure, the valve assembly is operable to automatically assume the closed position when the filter housing is detached from the collection jar.

[0011] According to another aspect of the disclosure, the valve assembly comprises a housing slide configured to close the base drain of the filter housing when the collection jar is detached from the filter housing.

[0012] According to another aspect of the disclosure, the valve assembly further comprises a housing slide biasing member configured to bias the housing slide toward the base drain.

[0013] According to another aspect of the disclosure, the valve assembly further comprises an elastomeric housing seal to provide a water-tight seal between the housing slide and the base drain when the valve assembly is in the closed position.

[0014] According to another aspect of the disclosure, the valve assembly comprises a jar slide configured to close the cover drain of the collection jar when the collection jar is detached from the filter housing.

[0015] According to another aspect of the disclosure, the valve assembly further comprises a jar slide biasing member configured to bias the jar slide toward the cover drain.

[0016] According to another aspect of the disclosure, the valve assembly further comprises an elastomeric jar seal to provide a water-tight seal between the jar slide and the cover drain when the valve assembly is in the closed position.

[0017] According to another aspect of the disclosure, a latch mechanism is configured to securely and releasably attach the filter housing to the collection jar.

[0018] According to another aspect of the disclosure, the latch mechanism is operable to permit detachment of the collection jar from the filter housing without disconnecting the filter housing from the breathing circuit.

[0019] According to another aspect of the disclosure, an impact pad is provided within the filter housing and operable to remove liquid from the flow of expiratory air.

[0020] According to another aspect of the disclosure, the impact pad is an electrostatic pad.

[0021] According to another aspect of the disclosure, the filter member is pleated.

[0022] According to another aspect of the disclosure, the respiratory filter and condensate

management apparatus for a breathing circuit may comprise a filter housing including a patient end and a ventilator end, the patient end having an air inlet port configured to receive a flow of expiratory air, and the ventilator end having an air outlet port configured to output the flow of expiratory air; a filter member provided within the filter housing and located in a flow path of the expiratory air between the air inlet port and the air outlet port; a collection jar removably attached to the filter housing, the collection jar including a patient end liquid reservoir operable to collect liquid formed by condensation in the flow of expiratory air within the filter housing when the collection jar is attached to the filter housing; and a patient end valve assembly configured to move to an open position when the filter housing is attached to the collection jar, and further configured to move to a closed position when the filter housing is detached from the collection jar.

[0023] According to another aspect of the disclosure, the filter housing further includes a base having a patient end base drain.

[0024] According to another aspect of the disclosure, the collection jar further includes a spill cover having a patient end cover drain, the patient end cover drain and the patient end base drain forming a patient end liquid passageway operable to provide fluid communication between the filter housing and the patient end liquid reservoir when the collection jar is attached to the filter housing.

[0025] According to another aspect of the disclosure, the patient end valve assembly is further configured to permit passage of the liquid formed by condensation in the flow of expiratory gas through the patient end liquid passageway when the patient end valve assembly is in an open position, and further configured to prevent passage of the liquid through the patient end liquid passageway when the patient end valve assembly is in a closed position.

[0026] According to another aspect of the disclosure, the collection jar further includes a ventilator end liquid reservoir operable to collect liquid formed by condensation in the flow of expiratory air within the filter housing when the collection jar is attached to the filter housing.

[0027] According to another aspect of the disclosure, a ventilator end valve assembly is configured to move to an open position when the filter housing is attached to the collection jar, and further configured to move to a closed position when the filter housing is detached from the collection jar.

[0028] There has thus been outlined certain embodiments of the disclosure in order that the detailed description thereof herein may be better understood, and in order that the present contribution to the art may be better appreciated. There are additional embodiments of the disclosure that will be described below and which form the subject matter of the claims appended hereto.

[0029] In this respect, before explaining at least one aspect of the respiratory filter and condensate management apparatus in detail, it is to be understood that the respiratory filter and condensate management apparatus is not limited in its application to the details of construction and to the arrangements of the components set forth in the following description or illustrated in the drawings. The respiratory filter and condensate management apparatus is capable of aspects in addition to those described, and of being practiced and carried out in various ways. Also, it is to be understood that the phraseology and terminology employed herein, as well as the abstract, are for the purpose of description and should not be regarded as limiting.

[0030] As such, those skilled in the art will appreciate that the conception upon which this disclosure is based may readily be utilized as a basis for the designing of other structures, methods, and systems for carrying out the several purposes of the respiratory filter and condensate management apparatus. It is important, therefore, that the claims be regarded as including such equivalent constructions insofar as they do not depart from the spirit and scope of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0031] In order that the disclosure may be readily understood, aspects of the respiratory filter and

condensate management apparatus are illustrated by way of examples in the accompanying drawings, in which like parts are referred to with like reference numerals throughout.

[0032] FIG. 1 is a perspective view of a self-sealing respiratory filter and condensate management apparatus for a breathing circuit according to an implementation of the present disclosure.

[0033] FIG. 2 is a cross-sectional side view of the respiratory filter and condensate management apparatus taken along line 2-2 of FIG. 1.

[0034] FIG. 3 is a cross-sectional side view of the respiratory filter and condensate management apparatus taken along line 3-3 of FIG. 1.

[0035] FIG. 4 is a cross-sectional perspective view of a ventilator-side self-closing valve assembly of the respiratory filter and condensate management apparatus in accordance with the present disclosure.

[0036] FIG. 5 is a perspective view of a jar slide of the valve assembly in accordance with the present disclosure.

[0037] FIG. 6 is a cross-sectional side view of the self-sealing respiratory filter and condensate management apparatus in a detached state.

[0038] FIG. 7 is a perspective view of a self-sealing respiratory filter and condensate management apparatus according to another implementation of the present disclosure.

[0039] FIG. 8 is a cross-sectional view from a side of the respiratory filter and condensate management apparatus of FIG. 7.

[0040] FIG. 9 is a cross-sectional view from a front of the respiratory filter and condensate management apparatus of FIG. 8.

[0041] FIG. 10a is a partial cross-sectional top perspective view of the ventilator-side valve assembly of the respiratory filter and condensate management apparatus of FIG. 9.

[0042] FIG. 10b is a partial cross-sectional bottom perspective view of the ventilator-side valve assembly of the respiratory filter and condensate management apparatus of FIG. 9.

[0043] FIG. 11 is a perspective cross-sectional view of the self-sealing respiratory filter and condensate management apparatus of FIG. 9.

[0044] FIG. 12 is a perspective view of a self-sealing respiratory filter and condensate management apparatus according to another implementation of the present disclosure.

[0045] FIG. 13 is a cross-sectional view from a side of the apparatus depicted in FIG. 12.

[0046] FIG. 14 is a cross-sectional view from a front of the apparatus depicted in FIG. 12.

[0047] FIG. 15 is a partial cross-sectional view of a self-sealing valve assembly of the apparatus depicted in FIG. 14.

[0048] FIG. 16 is a cross-sectional perspective view of the apparatus depicted in FIG. 14.

[0049] FIG. 17 is a cross-sectional view of a self-sealing respiratory filter and condensate management apparatus in an attached state according to another implementation of the present disclosure.

[0050] FIG. 18 is a cross-sectional view of the self-sealing respiratory filter and condensate management apparatus of FIG. 17 in a detached state.

DETAILED DESCRIPTION

[0051] FIGS. 1-6 illustrate a self-sealing respiratory filter and condensate management apparatus 100 for a breathing circuit in accordance with an implementation of the present disclosure. The respiratory filter and condensate management apparatus 100 comprises a filter housing 120 and a collection jar 140. The collection jar 140 is releasably attachable to the filter housing 120. The filter housing 120 defines a filter compartment, and the collection jar 140 defines a condensate compartment. The filter housing 120 includes base 121, an air inlet port 122, and an air outlet port 124. The air inlet port 122 is configured to connect to an expiratory limb of the breathing circuit for receiving a flow of expiratory air, and the air outlet port 124 is configured to connect to a ventilator either directly or via a breathing circuit tubing for outputting the flow of expiratory air. A self-closing valve assembly 150a,b is provided to regulate passage of condensate from the filter

compartment of the housing **120** to the condensate compartment of the collection jar **140** when the collection jar is attached to the filter housing, as will be discussed in greater detail below.

[0052] The respiratory filter and condensate management apparatus **100** may include one or more latch mechanisms **126** for securely and releasably attaching the filter housing **120** to the collection jar **140**. For instance, FIG. 2 depicts a pair of latch mechanisms **126** provided on opposite sides of the respiratory filter and condensate management apparatus **100**. Each latch mechanism **126** is operable to securely and releasably attach the collection jar **140** to the filter housing **120**. As illustrated in FIG. 2, the latch mechanism **126** comprises a lever arm **127**, a protrusion or catch **128** extending from the lever arm, and a corresponding receptacle or notch **129** configured to receive a portion of the protrusion or catch **128** in an engaged or locked position. The latch mechanism may include a spring tab, a spring lever, a spring bar, a spring latch, and a snap-fit, among others. The lever arm **127** may include a non-slip gripping surface, such as a textured or roughened surface, among others

[0053] The lever arm **127** may be pivotably or flexibly connected to the collection jar **140**. The receptacle or notch **129** is formed on the filter housing **120**. The latch mechanism **126** is operable to lock the collection jar **140** to the filter housing **120** and release the collection jar from the filter housing by respectively engaging and disengaging the catch **128** from the corresponding notch **129**. In another implementation, the lever arm may be pivotably or flexibly connected to the filter housing and the receptacle or notch may be formed on the collection jar, such that the latch mechanism is operable to lock the collection jar to the filter housing and release the collection jar from the filter housing by respectively engaging and disengaging a portion of the catch from the corresponding notch.

[0054] A filter member **131** is provided within the filter compartment of the housing **120** and is in fluid communication with the air inlet port **122** and the air outlet port **124** for filtering bacteria, viruses, medicament, and/or other waste material exhaled by a patient during respiration. More particularly, the filter member **131** is located in an air flow path from the air inlet port **122** to the air outlet port **124**. The filter member **131** may be made from material, such as micro-fiberglass, that is operable to catch bacteria, viruses, medicament, and/or other waste material, while still allowing the flow of air to pass through the filter member. For instance, the filter member **131** may be a high efficiency particulate air (HEPA) filter, or an ultra low particulate air (ULPA) filter, among others. In some implementations, the filter member **131** may be non-pleated, while in other implementations the filter member may be pleated to improve the efficiency of air filtration. The filter member **131** may also be operable to let gaseous vapor comprising small water droplets to pass through. In some aspects, the filter member may be treated with an antimicrobial agent.

[0055] As shown in FIGS. 2, 3 and 6, the filter member **131** is configured to fit within the filter compartment of the filter housing **120** such that the flow of air from the air inlet port **122** to the air outlet port **124** must pass through the filter member. Stated another way, the filter member **131** has a shape and size relative to a shape and size of the filter compartment of the housing **120** such that the flow of air from the air inlet port **122** to the air outlet port **124** is prevented from bypassing the filter member **131**. The filter member **131** may have a generally cuboidal shape, among others.

[0056] Heated and humidified air that enters the air inlet port **122** may produce condensate within the filter compartment of the housing **120** due to the presence of colder air within the filter compartment. Condensation or rainout may occur in a patient side **132** of the filter compartment before the air flow passes through the filter member **131**, and thus condensate that accumulates within the patient side **132** of the filter compartment may be dirty water. As previously described, the filter member **131** may be operable to let gaseous vapor comprising water droplets pass through. Accordingly, condensation or rainout may also occur in a ventilator side **134** of the filter compartment of the housing after the air flow passes through the filter member **131**, and thus condensate that accumulates within the ventilator side **134** of the filter compartment may be clean water.

[0057] When the filter housing **120** is attached to the collection jar **140**, the condensate compartment is located directly adjacent to and below the filter compartment and is configured to collect liquid water formed by condensation within the filter compartment. In particular, the condensate compartment of the collection jar **140** includes a first reservoir **142** operable to collect liquid formed by condensation within the patient side **132** of the filter compartment, and a second reservoir **144** operable to collect liquid formed by condensation within the ventilator side **134** of the filter compartment. A partition wall **145** may be provided within the condensate compartment between the first and second reservoirs **142**, **144** to separate the reservoirs and ensure the dirty water within the first reservoir **142** does not mix with the clean water in the second reservoir **144**. The collection jar **140** also includes a cover **141** to prevent liquid from spilling out of the reservoirs when the collection jar is detached from the filter housing.

[0058] A pre-filter impact pad **135** may also be provided within the patient side **132** of the filter compartment of the housing **120**. The impact pad **135** is operable to remove liquid particles from the humidified flow of air before the humidified air reaches the filter member **131**. For instance, such liquid particles removed from the air flow by the impact pad may include medicament particles. As shown in FIGS. **3** and **6**, the impact pad **135** is located between the filter member **131** and the air inlet port **122**. Thus, the humidified flow of air contacts the impact pad **135** before reaching the filter member **131**. The air flow may be diverted around the pad and/or through the pad toward the filter member in order to enhance impaction or catching of aerosol particles.

[0059] The impact pad **135** may be absorbent and therefore able to retain the liquid particles. Further, the impact pad may be electrostatically charged for treating the flow of air from the air inlet port **122**. For instance, the electrostatic impact pad may comprise a fibrous material imbued with an electrical charge during manufacturing. Thus, the electrostatic impact pad **135** is operable to attract and retain liquid particles in the air flow so as to prevent their further travel into the filter member **131**, thus prolonging the lifespan of the filter member.

[0060] The impact pad **135** may have generally a planar front surface positioned so as to directly face the air inlet port **122** in a direction perpendicular to the flow of air from the air inlet port **122**. Moreover, the impact pad **135** may be shaped and sized such that its cross-section is larger than a cross-section of the air inlet port **122**. The impact pad **135** is furthermore located in the direct flow path of air entering through the air inlet port **122**. In some aspects the impact pad may have a square or rectangular cross-sectional shape, among others.

[0061] The impact pad **135** may be mounted on a mounting frame **139** within the patient side **132** of the filter compartment of the housing **120** so that a clearance surrounding a periphery of the impact pad is formed for allowing the flow of air from the air inlet port to be diverted around the impact pad toward the filter member. The clearance defines a space or gap between the impact pad and an interior surface of the filter housing. The clearance allows the flow of air to be diverted around a top of the pad, a bottom of the pad, and/or the lateral sides of the pad. The impact pad **135** may be secured to the mounting frame **139** by a fastener, such as an adhesive or a clip, among others. According to other aspects, the impact pad may have a conical shape or a cylindrical shape, among others. The mounting frame **139** may also include at least one opening configured to allow the air flow to pass from the air inlet port **122** to the filter member **131** through the impact pad **135**.

[0062] During use, the electrostatic impact pad **135** causes the air to diffuse, slow down, and be diverted around the pad and/or flow through the pad toward the filter member. The impact pad therefore enhances impaction or catching of aerosol particles. The electrostatic charge on the impact pad **135** attracts liquid particles from the flow of air, thereby preventing the liquid particles from reaching the filter member **131**. Accordingly, the electrostatic impact pad is operable to scrub out aerosol from the flow of air from the air inlet port.

[0063] During use in patient respiration, liquid condensation that forms within the filter compartment of the housing **120** will drip downward toward the condensate compartment of the collection jar **140** due to the effect of gravity. More specifically, condensate within the patient side

132 of the filter compartment will pass downward through a first condensate or liquid passageway **136** and into the first reservoir **142** of the condensate compartment due to gravity. Similarly, condensate within the ventilator side **134** of the filter compartment will pass downward through a second condensate or liquid passageway **138** and into the second reservoir **144** of the condensate compartment due to gravity.

[0064] As depicted in FIG. 3, the first condensate passageway **136** extends through respective portions of the base **121** of the housing, as well as the cover **141** of the collection jar. Likewise, the second condensate passageway **138** extends through other respective portions of the base **121** of the housing and the cover **141** of the collection jar. Each of the first and second condensate passageways **136**, **138** are surrounded by a corresponding annular groove **148** formed in an outer surface of the jar cover. Each annular groove **148** is configured to receive a portion of an elastomeric seal **149**, such as a rubber O-ring, that provides a fluid tight seal between the housing base **121** and the jar cover **141** when the jar is attached to the housing. A first support member **146**, such as a column or pillar, is disposed within the collection jar and is axially aligned with the first condensate passage **136**. A second support member **147**, such as a column or pillar, is disposed within the collection jar and is axially aligned with the second condensate passage **138**.

[0065] When the collection jar **140** is attached to the filter housing **120**, a patient-side self-closing valve assembly **150a** controls the passage of condensate flowing from the patient side **132** of the filter compartment of the housing through the first condensate passageway **136** and into the first reservoir **142** of the condensate compartment of the collection jar. Similarly, when the collection jar **140** is attached to the filter housing **120**, a ventilator-side self-closing valve assembly **150b** controls the passage of condensate flowing from the ventilator side **134** of the filter compartment of the housing through the second condensate passageway **138** and into the second reservoir **144** of the condensate compartment of the collection jar. Each self-closing valve assembly **150a,b** is configured to move to an open position when the filter housing **120** is attached to the collection jar **140**, and further configured to move to a closed position when the filter housing is detached from the collection jar, as will be discussed further below.

[0066] An enlarged view of the ventilator-side self-closing valve assembly **150b** is depicted in FIG. 4 for illustration purposes since it should be appreciated that component parts of the patient-side self-closing valve assembly **150a** and the ventilator-side self-closing valve assembly **150b** are alike. With reference to FIG. 4, the ventilator-side self-closing valve assembly **150b** comprises a valve cap **160** and a housing slide **164** that are both disposed within the filter housing. The valve cap **160** may be fixedly secured to the housing or formed integrally with the housing. In some aspects, the patient-side self-closing valve assembly **150a** may be fixedly secured to the mounting frame **139** or formed integrally with the mounting frame. In other aspects, the patient-side self-closing valve assembly **150a** may be fixedly secured to the housing **120** or formed integrally with the housing.

[0067] The valve cap **160** includes a recess configured to receive a housing slide biasing member **168**, such as a compression spring. The housing slide biasing member **168** is configured to bias the housing slide **164** toward a closed position in which condensate is prevented from passing through the second condensate passageway **138**. In some implementations, the housing slide **164** may include an elastomeric housing seal **169**, such as a silicone washer, to ensure a water-tight seal with the base **121** of the housing **120** when in the closed position to prevent condensate from leaking out of the housing through the corresponding condensate passageway.

[0068] The ventilator-side self-closing valve assembly **150b** further comprises a jar slide **151** at least partly disposed within the collection jar **140**. As further illustrated in FIG. 5, the jar slide **151** includes a flange **152**, which may be annular, among other shapes. A base **153** extends from a first surface of the flange **152**, and a valve stem or jar stem **154** extends from a second surface of the flange **152**. The valve stem **154** includes at least one vane **155** extending laterally therefrom. As depicted in FIG. 5, the valve stem **154** includes a plurality of vanes **155** extending laterally

therefrom. In particular, the valve stem **154** includes four vanes **154** equally spaced apart to form a cross-section shaped as a cross or plus (+) sign. In other implementations, the valve stem may include more or less than four vanes, either equally or not equally spaced apart, and forming other shaped cross-sections. The valve stem **154** further includes a tip **156** configured to abut the housing slide **164** when in the open position.

[0069] The base **153** may be cylindrical, among other shapes, and includes a recess configured to receive a jar slide biasing member **158**, such as a compression spring. The jar slide **151** is supported by the corresponding support member protruding into the collection jar. For instance, FIG. 4 depicts the jar slide **151** is supported by the second support member **147** protruding into the collection jar. The jar slide biasing member **158** is configured to bias the jar slide **151** toward a closed position in which condensate is prevented from spilling out of the second reservoir **144** through the second condensate passageway **138**. In some implementations, the jar slide **151** may include an elastomeric jar seal **159**, such as a silicone washer, to ensure a water-tight seal with the cover **141** of the collection jar **140** when in the closed position to prevent condensate from leaking out of the collection jar through the second condensate passageway **138**.

[0070] In operation, when the collection jar **140** is attached to the filter housing **120**, as depicted in FIGS. 3 and 4, the tip **156** of the jar slide **151** contacts the housing slide **164** such that both the jar slide and the housing slide are respectively urged against the biasing force of the jar slide biasing member **158** and the housing slide biasing member **168** toward respective open positions in which liquid condensate is allowed to pass through the corresponding condensate passageway. A channel defined by adjacent vanes **155** of the valve stem **154** allows the condensate to pass through the condensate passageway when the valve stem is inserted into the condensate passageway. Thus, when the collection jar **140** is attached to the filter housing **120**, each valve assembly **150a,b** automatically assumes the open position to allow condensate to pass from the filter housing through the corresponding condensate passageway and into the collection jar.

[0071] When the collection jar **140** is detached from the filter housing **120**, as depicted in FIG. 6, the tip **156** of the jar slide **151** no longer contacts the housing slide **164**, such that both the jar slide and the housing slide are respectively urged by the biasing force of the jar slide biasing member **158** and the housing slide biasing member **168** toward the closed position in which the corresponding condensate passage is blocked, thus preventing liquid condensate from passing therethrough. As described above, in the closed position, the flange **152** of the jar slide sealingly abuts an interior surface of a portion of the collection jar cover **141** surrounding the corresponding condensate passageway to block it, and thus preventing condensate from spilling out of the collection jar through the condensate passageway. The elastomeric jar seal **159** may be disposed between the flange **152** and the interior surface of the collection jar to ensure a fluid tight seal when the valve assembly is in the closed position.

[0072] Similarly, the housing slide **164** sealingly abuts an interior surface of a portion of the filter housing base **121** surrounding the corresponding condensate passageway to block it, and thus preventing condensate from leaking out of the filter housing through the condensate passageway. The elastomeric housing seal **169** may be disposed between the housing slide **164** and the interior surface of the filter housing base to ensure a fluid tight seal. Thus, when the collection jar **140** is detached from the filter housing **120**, each valve assembly **150a,b** automatically moves to the closed position, thus obstructing the corresponding condensate passageway, and therefore preventing condensate from leaking out of both the filter housing and the collection jar through the corresponding condensate passageway.

[0073] FIGS. 7-11 illustrate a self-sealing respiratory filter and condensate management apparatus **200** for a breathing circuit in accordance with another implementation of the present disclosure. The respiratory filter and condensate management apparatus **200** comprises a filter housing **220** and a collection jar **240**. The collection jar **240** is releasably attachable to the filter housing **220**. The filter housing **220** defines a filter compartment, and the collection jar **240** defines a condensate

compartment. The filter housing **220** includes a base **221**, an air inlet port **222**, and an air outlet port **224**. The air inlet port **222** is configured to connect to an expiratory limb of the breathing circuit, and the air outlet port **224** is configured to connect to a ventilator either directly or via a breathing circuit tubing. A self-closing valve assembly **250a,b** is provided to regulate passage of condensate from the filter compartment of the housing **220** to the condensate compartment of the collection jar **240** when the collection jar is attached to the filter housing.

[0074] The respiratory filter and condensate management apparatus **200** may include one or more latch mechanisms **226** for securely and releasably attaching the filter housing **220** to the collection jar **240**. For instance, FIG. **8** depicts a pair of latch mechanisms **226** provided on opposite sides of the respiratory filter and condensate management apparatus **200**. Each latch mechanism **226** is operable to securely and releasably attach the collection jar **240** to the filter housing **220**. As illustrated in FIG. **8**, the latch mechanism **226** comprises a lever arm **227**, a protrusion or catch **228** extending from the lever arm, and a corresponding receptacle or notch **229** configured to receive a portion of the protrusion or catch **228** in an engaged or locked position. The latch mechanism may include a spring tab, a spring lever, a spring bar, a spring latch, and a snap-fit, among other configurations. The lever arm **227** may include a non-slip gripping surface, such as a textured or roughened surface, among others.

[0075] The lever arm **227** may be pivotably or flexibly connected to the collection jar **240**. The receptacle or notch **229** is formed on the filter housing **220**. The latch mechanism **226** is operable to lock the collection jar **240** to the filter housing **220** and release the collection jar from the filter housing by respectively engaging and disengaging the catch **228** from the corresponding notch **229**. In another implementation, the lever arm may be pivotably or flexibly connected to the filter housing and the receptacle or notch may be formed on the collection jar, such that the latch mechanism is operable to lock the collection jar to the filter housing and release the collection jar from the filter housing by respectively engaging and disengaging a portion of the catch from the corresponding notch.

[0076] A filter member **231** is provided within the filter compartment of the housing **220** and is in fluid communication with the air inlet port **222** and the air outlet port **224** for filtering bacteria, viruses, medicament, and/or other waste material exhaled by a patient during respiration. More particularly, the filter member **231** is located in an air flow path from the air inlet port **222** to the air outlet port **224**. The filter member **231** may be made from material, such as micro-fiberglass, that is operable to catch bacteria, viruses, medicament, and/or other waste material, while still allowing the flow of air to pass through the filter member. For instance, the filter member **231** may be a high efficiency particulate air (HEPA) filter, or an ultra low particulate air (ULPA) filter, among others. In some implementations, the filter member **231** may be non-pleated, while in other implementations the filter member may be pleated to improve the efficiency of air filtration. The filter member **231** may also be operable to let gaseous vapor comprising small water droplets to pass through. In some aspects, the filter member may be treated with an antimicrobial agent.

[0077] As shown in FIGS. **8**, **9** and **11**, the filter member **231** is configured to fit within the filter compartment of the filter housing **220** such that the flow of air from the air inlet port **222** to the air outlet port **224** must pass through the filter member. Stated another way, the filter member **231** has a shape and size relative to a shape and size of the filter compartment of the housing **220** such that the flow of air from the air inlet port **222** to the air outlet port **224** is prevented from bypassing the filter member **231**. The filter member **231** may have a generally cuboidal shape, among others.

[0078] Heated and humidified air that enters the air inlet port **222** may produce condensate within the filter compartment of the housing **220** due to the presence of colder air within the filter compartment. Condensation or rainout may occur in a patient side **232** of the filter compartment before the air flow passes through the filter member **231**, and thus condensate that accumulates within the patient side **232** of the filter compartment may be dirty water. As previously described, the filter member **231** may be operable to let gaseous vapor comprising water droplets pass

through. Accordingly, condensation or rainout may also occur in a ventilator side **234** of the filter compartment of the housing after the air flow passes through the filter member **231**, and thus condensate that accumulates within the ventilator side **234** of the filter compartment may be clean water.

[0079] When the filter housing **220** is attached to the collection jar **240**, the condensate compartment is located directly adjacent to and below the filter compartment and is configured to collect liquid water formed by condensation within the filter compartment. In particular, the condensate compartment of the collection jar **240** includes a first reservoir **242** operable to collect liquid formed by condensation within the patient side **232** of the filter compartment, and a second reservoir **244** operable to collect liquid formed by condensation within the ventilator side **232** of the filter compartment. A partition wall **245** may be provided within the condensate compartment between the first and second reservoirs **242**, **244** to separate the reservoirs and ensure the dirty water within the first reservoir **242** does not mix with the clean water in the second reservoir **244**. The collection jar **240** also includes a cover **241** to prevent liquid from spilling out of the reservoirs when the collection jar is detached from the filter housing.

[0080] A pre-filter impact pad **235** may also be provided within the patient side **232** of the filter compartment of the housing **220**. The impact pad **235** is operable to remove liquid particles from the humidified flow of air before the humidified air reaches the filter member **231**. For instance, such liquid particles removed from the air flow by the impact pad may include medicament particles. As shown in FIGS. **9** and **11**, the impact pad **235** is located between the filter member **231** and the air inlet port **222**. Thus, the humidified flow of air contacts the impact pad **235** before reaching the filter member **231**. The air flow may be diverted around the pad and/or through the pad toward the filter member in order to enhance impaction or catching of aerosol particles.

[0081] The impact pad **235** may be absorbent and therefore able to retain the liquid particles. Further, the impact pad may be electrostatically charged for treating the flow of air from the air inlet port **222**. For instance, the electrostatic impact pad may comprise a fibrous material imbued with an electrical charge during manufacturing. Thus, the electrostatic impact pad **235** is operable to attract and retain liquid particles in the air flow so as to prevent their further travel into the filter member **231**, thus prolonging the lifespan of the filter member.

[0082] The impact pad **235** may have generally a planar front surface positioned so as to directly face the air inlet port **222** in a direction perpendicular to the flow of air from the air inlet port **222**. Moreover, the impact pad **235** may be shaped and sized such that its cross-section is larger than a cross-section of the air inlet port **222**. The impact pad **235** is furthermore located in the direct flow path of air entering through the air inlet port **222**. In some aspects the impact pad may have a square or rectangular cross-sectional shape, among others.

[0083] The impact pad **235** may be mounted on a mounting frame **239** within the patient side **232** of the filter compartment of the housing **220** so that a clearance surrounding a periphery of the impact pad is formed for allowing the flow of air from the air inlet port to be diverted around the impact pad toward the filter member. The clearance defines a space or gap between the impact pad and an interior surface of the filter housing. The clearance allows the flow of air to be diverted around a top of the pad, a bottom of the pad, and/or the lateral sides of the pad. The impact pad **235** may be secured to the mounting frame **239** by a fastener, such as an adhesive or a clip, among others. According to other aspects, the impact pad may have a conical shape or a cylindrical shape, among others. The mounting frame **239** may also include at least one opening configured to allow the air flow to pass from the air inlet port **222** to the filter member **231** through the impact pad **235**.

[0084] Moreover, during use the electrostatic impact pad **235** causes the air to diffuse, slow down, and be diverted around the pad and/or flow through the pad toward the filter member. The impact pad therefore enhances impaction or catching of aerosol particles. The electrostatic charge on the impact pad **235** attracts liquid particles from the flow of air, thereby preventing the liquid particles from reaching the filter member **231**. Accordingly, the electrostatic impact pad is operable to scrub

out aerosol from the flow of air from the air inlet port.

[0085] During use in patient respiration, liquid condensation that forms within the filter compartment of the housing **220** will drip downward toward the condensate compartment of the collection jar **240** due to the effect of gravity. More specifically, condensate within the patient side **232** of the filter compartment will pass downward through a first condensate or liquid passageway **236** and into the first reservoir **242** of the condensate compartment due to gravity. Similarly, condensate within the ventilator side **234** of the filter compartment will pass downward through a second condensate or liquid passageway **238** and into the second reservoir **244** of the condensate compartment due to gravity.

[0086] The first condensate passageway **236** extends through respective portions of the base **221** of the housing and the cover **241** of the collection jar. Likewise, the second condensate passageway **238** extends through other respective portions of the base **221** of the housing and the cover **241** of the collection jar. In some aspects, one or more apertures **223** may be provided in the filter housing base **221** and configured to provide fluid communication between the filter compartment and the corresponding condensate passage when the respective valve assembly is in the open position. Similarly, one or more apertures **243** may be provided in the jar cover **241** and configured to provide fluid communication between the condensate collection compartment and the corresponding condensate passage when the respective valve assembly is in the open position.

[0087] Each of the first and second condensate passageways **236**, **238** are surrounded by a corresponding annular groove **248** formed in an outer surface of the jar cover. Each annular groove **248** is configured to receive a portion of an elastomeric seal **249**, such as a rubber O-ring, that provides a fluid tight seal between the housing base **221** and the jar cover **241** when the jar is attached to the housing. A first support member **246**, such as a column or pillar, is disposed within the collection jar and is axially aligned with the first condensate passage **236**. A second support member **247**, such as a column or pillar, is disposed within the collection jar and is axially aligned with the second condensate passage **238**.

[0088] When the collection jar **240** is attached to the filter housing **220**, a patient-side self-closing valve assembly **250a** controls the passage of condensate flowing from the patient side **232** of the filter compartment of the housing through the first condensate passageway **236** and into the first reservoir **242** of the condensate compartment of the collection jar. Similarly, when the collection jar **240** is attached to the filter housing **220**, a ventilator-side self-closing valve assembly **250b** controls the passage of condensate flowing from the ventilator side **234** of the filter compartment of the housing through the second condensate passageway **238** and into the second reservoir **244** of the condensate compartment of the collection jar. Each self-closing valve assembly **250a,b** is configured to move to an open position when the filter housing **220** is attached to the collection jar **240**, and further configured to move to a closed position when the filter housing is detached from the collection jar, as will be discussed further below.

[0089] An enlarged view of the ventilator-side self-closing valve assembly **250b** is depicted in FIGS. **10a** and **10b** for illustration purposes since it should be appreciated that component parts of the patient-side self-closing valve assembly **250a** and the ventilator-side self-closing valve assembly **250b** are alike. The ventilator-side self-closing valve assembly **250b** comprises a valve cap **260** and a housing slide **261** that are both disposed within the filter housing. The valve cap **260** may be fixedly secured to the housing or formed integrally with the housing. In some aspects, the patient-side self-closing valve assembly **250a** may be fixedly secured to the mounting frame **239** or formed integrally with the mounting frame. In other aspects, the patient-side self-closing valve assembly **250a** may be fixedly secured to the housing **220** or formed integrally with the housing.

[0090] The jar slide **251** includes a flange **252**, which may be annular, among other shapes. A valve stem or jar stem **254** extends from a surface of the flange **252** and includes a tip **256** configured to abut the housing slide **261** when in the open position. The flange **252** is configured to abut a jar slide biasing member **258**, such as a compression spring. The jar slide **251** is supported by the

corresponding support member protruding into the collection jar. For instance, FIGS. **10a** and **10b** depict the jar slide **251** is supported by the second support member **247** protruding into the collection jar. The jar slide biasing member **258** is configured to bias the jar slide **251** toward a closed position in which condensate is prevented from spilling out of the second reservoir **244** through the second condensate passageway **238**. In some implementations, the jar slide **251** may include an elastomeric jar seal **259**, such as a silicone washer, to ensure a water-tight seal over the at least one aperture **243** in the jar cover **241** when in the closed position to prevent condensate from leaking out of the collection jar through the second condensate passageway **238**.

[0091] The housing slide **261** includes a flange **262**, which may be annular, among other shapes. A base **263** extends from a first surface of the flange **262**, and a housing stem **265** extends from a second surface of the flange **262**. The housing stem **265** includes a tip **266** having a recess **267** configured to receive and abut the jar slide tip **256** when in the open position. The base **263** may be cylindrical, among other shapes, and includes a recess configured to receive a housing slide biasing member **268**, such as a compression spring. The valve cap **260** includes a recess configured to receive base **263** and the housing slide biasing member **268**. The housing slide biasing member **268** is configured to bias the housing slide **261** toward a closed position in which condensate is prevented from passing through the second condensate passageway **238**. In some implementations, the housing slide **261** may include an elastomeric housing seal **269**, such as a silicone washer, to ensure a water-tight seal over the at least one aperture **223** in the housing base **221** when in the closed position to prevent condensate from leaking out of the housing through the second condensate passageway **238**.

[0092] In operation, when the collection jar **240** is attached to the filter housing **220**, the tip **256** of the jar slide **251** contacts the housing slide **261** such that both the jar slide and the housing slide are respectively urged against the biasing force of the jar slide biasing member **258** and the housing slide biasing member **268** toward an open position in which liquid condensate is allowed to pass through the corresponding condensate passageway via the respective apertures **223**, **243**. Thus, when the collection jar **240** is attached to the filter housing **220**, each valve assembly **250a,b** automatically moves to the open position to allow condensate to pass from the filter housing through the corresponding condensate passageway and into the collection jar.

[0093] When the collection jar **240** is detached from the filter housing **220**, the tip **256** of the jar slide **251** no longer contacts the housing slide **261** such that both the jar slide and the housing slide are respectively urged by the biasing force of the jar slide biasing member **258** and the housing slide biasing member **268** toward the closed position in which the at least one aperture **223**, **243** of the corresponding condensate passage is blocked, thus preventing liquid from passing therethrough.

[0094] Stated another way, in the closed position, the flange **252** of the jar slide sealingly covers the at least one aperture **243** in the jar cover **241** to block the corresponding condensate passageway and thus prevent condensate from spilling out of the collection jar through the condensate passageway. The elastomeric jar seal **259** may be disposed between the flange **252** and the interior surface of the collection jar to ensure a fluid tight seal when the valve assembly is in the closed position. Similarly, the housing slide **261** sealingly covers the at least one aperture **223** in the filter housing base **221** to block the corresponding condensate passageway and thus prevent condensate from leaking out of the filter housing through the condensate passageway. The elastomeric housing seal **269** may be disposed between the flange **262** of the housing slide and the interior surface of the filter housing base to ensure a fluid tight seal. Thus, when the collection jar **240** is detached from the filter housing **220**, each valve assembly **250a,b** automatically moves to the closed position to obstruct the corresponding condensate passageway, and therefore prevent condensate from leaking out of both the filter housing and the collection jar through the corresponding condensate passageway.

[0095] FIGS. **12-16** illustrate a self-sealing respiratory filter and condensate management apparatus **300** for a breathing circuit in accordance with another implementation of the present disclosure.

The respiratory filter and condensate management apparatus **300** comprises a filter housing **320** and a collection jar **340**. The collection jar **340** is releasably attachable to the filter housing **320**. The filter housing **320** defines a filter compartment, and the collection jar **340** defines a condensate compartment. The filter housing **320** includes a base **321**, an air inlet port **322**, and an air outlet port **324**. The air inlet port **322** is configured to connect to an expiratory limb of the breathing circuit, and the air outlet port **324** is configured to connect to a ventilator either directly or via a breathing circuit tube. A self-closing valve assembly **350a,b** is provided to regulate passage of condensate from the filter compartment of the housing **320** to the condensate compartment of the collection jar **340** when the collection jar is attached to the filter housing.

[0096] The respiratory filter and condensate management apparatus **300** may include one or more latch mechanisms **326** for securely and releasably attaching the filter housing **320** to the collection jar **340**. For instance, a pair of latch mechanisms **326** may be provided on opposite sides of the respiratory filter and condensate management apparatus **300**. Each latch mechanism **326** is operable to securely and releasably attach the collection jar **340** to the filter housing **320**. As illustrated in FIG. **13**, the latch mechanism **326** comprises a lever arm **327**, a protrusion or catch **328** extending from the lever arm, and a corresponding receptacle or notch **329** configured to receive a portion of the protrusion or catch **328** in an engaged or locked position. The latch mechanism may include a spring tab, a spring lever, a spring bar, a spring latch, and a snap-fit, among other configurations. The lever arm **327** may include a non-slip gripping surface, such as a textured or roughened surface, among others.

[0097] In the implementation shown in FIG. **13**, the lever arm **327** is pivotably or flexibly connected to the collection jar **340**, and the receptacle or notch **329** is formed on the filter housing **320**. The latch mechanism **326** is operable to lock the collection jar **340** to the filter housing **320** and release the collection jar from the filter housing by respectively engaging and disengaging the catch **328** from the corresponding notch **329**. In another implementation, the lever arm may be pivotably or flexibly connected to the filter housing and the receptacle or notch may be formed on the collection jar, such that the latch mechanism is operable to lock the collection jar to the filter housing and release the collection jar from the filter housing by respectively engaging and disengaging a portion of the catch from the corresponding notch.

[0098] A filter member **331** is provided within the filter compartment of the housing **320** and is in fluid communication with the air inlet port **322** and the air outlet port **324** for filtering bacteria, viruses, medicament, and/or other waste material exhaled by a patient during respiration. More particularly, the filter member **331** is located in an air flow path from the air inlet port **322** to the air outlet port **324**. The filter member **331** may be made from material, such as micro-fiberglass, that is operable to catch bacteria, viruses, medicament, and/or other waste material, while still allowing the flow of air to pass through the filter member. For instance, the filter member **331** may be a high efficiency particulate air (HEPA) filter, or an ultra low particulate air (ULPA) filter, among others. In some implementations, the filter member **331** may be non-pleated, while in other implementations the filter member may be pleated to improve the efficiency of air filtration. The filter member **331** may also be operable to let gaseous vapor comprising small water droplets to pass through. In some aspects, the filter member may be treated with an antimicrobial agent.

[0099] As shown in FIGS. **13**, **14** and **16**, the filter member **331** is configured to fit within the filter compartment of the filter housing **320** such that the flow of air from the air inlet port **322** to the air outlet port **324** must pass through the filter member. Stated another way, the filter member **331** has a shape and size relative to a shape and size of the filter compartment of the housing **320** such that the flow of air from the air inlet port **322** to the air outlet port **324** is prevented from bypassing the filter member **331**. The filter member **331** may have a generally cuboidal shape, among others.

[0100] Heated and humidified air that enters the air inlet port **322** may produce condensate within the filter compartment of the housing **320** due to the presence of colder air within the filter compartment. Condensation or rainout may occur in a patient side **332** of the filter compartment

before the air flow passes through the filter member **331**, and thus condensate that accumulates within the patient side **332** of the filter compartment may be dirty water. As previously described, the filter member **331** may be operable to let gaseous vapor comprising water droplets pass through. Accordingly, condensation or rainout may also occur in a ventilator side **334** of the filter compartment of the housing after the air flow passes through the filter member **331**, and thus condensate that accumulates within the ventilator side **334** of the filter compartment may be clean water.

[0101] When the filter housing **320** is attached to the collection jar **340**, the condensate compartment is located directly adjacent to and below the filter compartment and is configured to collect liquid water formed by condensation within the filter compartment. In particular, the condensate compartment of the collection jar **340** includes a first reservoir **342** operable to collect liquid formed by condensation within the patient side **332** of the filter compartment, and a second reservoir **344** operable to collect liquid formed by condensation within the ventilator side **332** of the filter compartment. A partition wall **345** may be provided within the condensate compartment between the first and second reservoirs **342**, **344** to separate the reservoirs and ensure the dirty water within the first reservoir **342** does not mix with the clean water in the second reservoir **344**.

[0102] During use in patient respiration, liquid condensation that forms within the filter compartment of the housing **320** will drip downward toward the condensate compartment of the collection jar **340** due to the effect of gravity. More specifically, condensate within the patient side **332** of the filter compartment will pass downward through a first condensate or liquid passageway **336** and into the first reservoir **342** of the condensate compartment due to gravity. Similarly, condensate within the ventilator side **334** of the filter compartment will pass downward through a second condensate or liquid passageway **338** and into the second reservoir **344** of the condensate compartment due to gravity. As depicted in FIG. 15, the first condensate passageway **336** extends through the base **321** of the housing. Likewise, the second condensate passageway **338** extends through the base **321** of the housing. A first support member **346**, such as a column or pillar, is disposed within the collection jar and is axially aligned with the first condensate passage **336**. A second support member **347**, such as a column or pillar, is disposed within the collection jar and is axially aligned with the second condensate passage **338**.

[0103] When the collection jar **340** is attached to the filter housing **320**, a patient-side self-closing valve assembly **350a** controls the passage of condensate flowing from the patient side **332** of the filter compartment of the housing through the first condensate passageway **336** and into the first reservoir **342** of the condensate compartment of the collection jar. Similarly, when the collection jar **340** is attached to the filter housing **320**, a ventilator-side self-closing valve assembly **350b** controls the passage of condensate flowing from the ventilator side **334** of the filter compartment of the housing through the second condensate passageway **338** and into the second reservoir **344** of the condensate compartment of the collection jar. Each self-closing valve assembly **350a,b** is configured to be in an open position when the filter housing **320** is attached to the collection jar **340**, and further configured to be in a closed position when the filter housing is detached from the collection jar.

[0104] When the collection jar **340** is attached to the filter housing **320**, a first self-closing valve assembly **250a** controls the passage of condensate flowing from the patient side **332** of the filter compartment **330** through the first condensate passageway **336** and into the first reservoir **342** of the condensate compartment. An enlarged view of the patient-side self-closing valve assembly **350a** is depicted in FIG. 15 for illustration purposes since it should be appreciated that component parts of the patient-side self-closing valve assembly **350a** and the ventilator-side self-closing valve assembly **350b** are alike.

[0105] The valve assembly **350a** includes a jar slide **351** comprising a base **353** attached to a valve stem or jar stem **354**. The valve stem **354** includes a proximal end having an enlarged valve head **355**, and a distal end having a valve tip **356**. A distal end of the base **353** includes a flange **352**,

which may be annular, among other shapes. The base **353** also includes at least one longitudinal valve slot **357** configured to allow liquid condensate to pass therethrough when the valve assembly is in the open position. A valve wall **325** protrudes from a bottom surface of the base **321** of the filter housing. The valve wall **325** may be cylindrical, among other shapes, and defines a recess configured to receive a jar slide biasing member **358** and a portion of the base **353** of the jar slide. The jar slide biasing member **358**, such as a compression spring, is disposed between the base **321** of the housing and the base **353** of the jar slide. The base **353** of the jar slide also includes a central longitudinal through-hole **360** in which a portion of the valve stem **354** is received. The tip **356** of the valve stem protrudes from the distal end of the base **353** of the jar slide and is operable to abut the corresponding support member protruding into the collection jar.

[0106] In FIG. **15**, for instance, the valve tip **356** abuts the first support member **346** protruding into the collection jar when the valve assembly **350a** is in an open position. The valve head **355** at the proximal end of the valve stem is configured to allow condensate to pass through the corresponding condensate passageway in the base **321** of the housing when the valve assembly is in the open position. Further, the valve head **355** is configured to block condensate from passing through the corresponding condensate passageway in the base **321** of the housing when the valve assembly is in the closed position. In some implementations, the jar slide **351** may include an elastomeric jar seal **359**, such as a silicone washer, to ensure a water-tight seal between the valve head **355** the corresponding condensate passageway when in the closed position to prevent condensate from leaking out of the housing **320**.

[0107] The valve tip **356** extends past a surface of the flange **352** and is operable to abut the corresponding support member **346** when in the open position. The base **353** is configured to abut the jar slide biasing member **358** such that the jar slide biasing member **358** is operable to bias the jar slide **351** toward a closed position in which condensate is prevented from dripping out of the housing **320** through the respective condensate passageway. Thus, in operation, when the collection jar **340** is attached to the filter housing **320**, the valve tip **356** of the jar slide **351** contacts the corresponding support member **346**, **347** such that the jar slide **351** is urged against the biasing force of the jar slide biasing member **358** toward an open position in which liquid condensate is allowed to pass through the corresponding condensate passageways **336**, **338**. Accordingly, when the collection jar **340** is attached to the filter housing **320**, each valve assembly **350a,b** automatically moves to the open position to allow condensate to pass from the filter housing through the corresponding condensate passageway and into the collection jar.

[0108] When the collection jar **340** is detached from the filter housing **320**, the valve tip **356** no longer contacts the corresponding support member, such that the jar slide **351** is urged by the biasing force of the jar slide biasing member **358** toward the closed position in which the corresponding condensate passage is blocked, thus preventing liquid condensate from passing therethrough. Stated another way, in the closed position, the enlarged valve head **355** sealingly covers the corresponding condensate passageway in the base **321** of the housing **320** to block the passageway, and thus prevent condensate from dripping out of the filter housing. Accordingly, when the collection jar **340** is detached from the filter housing **320**, each valve assembly **350a,b** automatically moves to the closed position to obstruct the corresponding condensate passageway, and therefore prevent condensate from leaking out of the filter housing through the corresponding condensate passageway.

[0109] FIGS. **17** and **18** illustrate a self-sealing respiratory filter and condensate management apparatus **400** for a breathing circuit in accordance with another implementation of the present disclosure. Similar to the apparatus **100** discussed in detail above, the respiratory filter and condensate management apparatus **400** comprises a filter housing **420** and a collection jar **440**. The collection jar **440** is releasably attachable to the filter housing **420**. The filter housing **420** defines a filter compartment, and the collection jar **440** defines a condensate compartment. The filter housing **420** includes base **421**, an air inlet port **422**, and an air outlet port **424**. The air inlet port **422** is

configured to connect to an expiratory limb of the breathing circuit, and the air outlet port **424** is configured to connect to a ventilator either directly or via a breathing circuit tubing. A self-closing valve assembly **450a,b** is provided to regulate passage of condensate from the filter compartment of the housing **420** to the condensate compartment of the collection jar **440** when the collection jar is attached to the filter housing.

[0110] The respiratory filter and condensate management apparatus **400** may include one or more latch mechanisms comprising one or more components or characteristics of any of the other latch mechanisms previously described above for securely and releasably attaching the filter housing **420** to the collection jar **440**.

[0111] A filter member **431** is provided within the filter compartment of the housing **420** and is in fluid communication with the air inlet port **422** and the air outlet port **424** for filtering bacteria, viruses, medicament, and/or other waste material exhaled by a patient during respiration. The filter member **431** may comprise one or more components or characteristics of any of the other filter members previously described above.

[0112] Heated and humidified air that enters the air inlet port **422** may produce condensate within the filter compartment of the housing **420** due to the presence of colder air within the filter compartment. Condensation or rainout may occur in a patient side **432** of the filter compartment before the air flow passes through the filter member **431**, and thus condensate that accumulates within the patient side **432** of the filter compartment may be dirty water. As previously described, the filter member **431** may be operable to let gaseous vapor comprising water droplets pass through. Accordingly, condensation or rainout may also occur in a ventilator side **434** of the filter compartment of the housing after the air flow passes through the filter member **431**, and thus condensate that accumulates within the ventilator side **434** of the filter compartment may be clean water.

[0113] When the filter housing **420** is attached to the collection jar **440**, the condensate compartment is located directly adjacent to and below the filter compartment and is configured to collect liquid water formed by condensation within the filter compartment. In particular, the condensate compartment of the collection jar **440** includes a first reservoir **442** operable to collect liquid formed by condensation within the patient side **432** of the filter compartment, and a second reservoir **444** operable to collect liquid formed by condensation within the ventilator side **434** of the filter compartment. A partition wall **445** may be provided within the condensate compartment between the first and second reservoirs **442**, **444** to separate the reservoirs and ensure the dirty water within the first reservoir **442** does not mix with the clean water in the second reservoir **444**. The collection jar **440** also includes a cover **441** to prevent liquid from spilling out of the reservoirs when the collection jar is detached from the filter housing.

[0114] During use in patient respiration, liquid condensation that forms within the filter compartment of the housing **420** will drip downward toward the condensate compartment of the collection jar **440** due to the effect of gravity. More specifically, condensate within the patient side **432** of the filter compartment will pass downward through a first condensate or liquid passageway **436** and into the first reservoir **442** of the condensate compartment due to gravity. Similarly, condensate within the ventilator side **434** of the filter compartment will pass downward through a second condensate or liquid passageway **438** and into the second reservoir **444** of the condensate compartment due to gravity.

[0115] The first condensate passageway **436** extends through respective portions of the base **421** of the housing, as well as the cover **441** of the collection jar. Likewise, the second condensate passageway **438** extends through other respective portions of the base **421** of the housing and the cover **441** of the collection jar. Each of the first and second condensate passageways **436**, **438** are surrounded by a corresponding annular groove **448** formed in an outer surface of the jar cover. Each annular groove **448** is configured to receive a portion of an elastomeric seal **449**, such as a rubber O-ring, that provides a fluid tight seal between the housing base **421** and the jar cover **441**.

when the jar is attached to the housing. A first support member **446**, such as a column or pillar, is disposed within the collection jar and is axially aligned with the first condensate passage **436**. A second support member **447**, such as a column or pillar, is disposed within the collection jar and is axially aligned with the second condensate passage **438**.

[0116] When the collection jar **440** is attached to the filter housing **420**, as shown in FIG. **17**, a patient-side self-closing valve assembly **450a** controls the passage of condensate flowing from the patient side **432** of the filter compartment of the housing through the first condensate passageway **436** and into the first reservoir **442** of the condensate compartment of the collection jar. Similarly, when the collection jar **440** is attached to the filter housing **420**, a ventilator-side self-closing valve assembly **450b** controls the passage of condensate flowing from the ventilator side **434** of the filter compartment of the housing through the second condensate passageway **438** and into the second reservoir **444** of the condensate compartment of the collection jar. Each self-closing valve assembly **450a,b** is configured to move to an open position when the filter housing **420** is attached to the collection jar **440**, and further configured to move to a closed position when the filter housing is detached from the collection jar.

[0117] Each self-closing valve assembly **450a,b** may comprise one or more components or characteristics of any of the other self-closing valve assemblies previously described above. In particular, the ventilator-side self-closing valve assembly **450b** comprises a valve cap **460** and a housing slide **464** that are both disposed within the filter housing. The valve cap **460** may be fixedly secured to the housing or formed integrally with the housing. In some aspects, the patient-side self-closing valve assembly **450a** may similarly be fixedly secured to the housing or formed integrally with the mounting frame.

[0118] The valve cap **460** includes a recess configured to receive a housing slide biasing member **468**, such as a compression spring. The housing slide biasing member **468** is configured to bias the housing slide **464** toward a closed position in which condensate is prevented from passing through the second condensate passageway **438**. In some implementations, the housing slide **464** may include an elastomeric housing seal **469**, such as a silicone washer, to ensure a water-tight seal with the base **421** of the housing **420** when in the closed position to prevent condensate from leaking out of the housing through the corresponding condensate passageway.

[0119] The ventilator-side self-closing valve assembly **450b** further comprises a jar slide **451** at least partly disposed within the collection jar **440**. The jar slide **451** includes a flange **452**, which may be annular, among other shapes. A base **453** extends from a first surface of the flange **452**, and a valve stem or jar stem extends from a second surface of the flange **452**. The valve stem includes at least one vane **455** extending laterally therefrom. The valve stem may include a plurality of vanes **455** extending laterally therefrom. The valve stem further includes a tip **456** configured to abut the housing slide **464** when in the open position, as shown in FIG. **17**.

[0120] The base **453** may be cylindrical, among other shapes, and includes a recess configured to receive a jar slide biasing member **458**, such as a compression spring. The jar slide **451** is supported by the corresponding support member protruding into the collection jar. The jar slide biasing member **458** is configured to bias the jar slide **451** toward a closed position in which condensate is prevented from spilling out of the second reservoir **444** through the second condensate passageway **438**. In some implementations, the jar slide **451** may include an elastomeric jar seal **459**, such as a silicone washer, to ensure a water-tight seal with the cover **441** of the collection jar **440** when in the closed position to prevent condensate from leaking out of the collection jar through the second condensate passageway **438**.

[0121] In operation, when the collection jar **440** is attached to the filter housing **420**, as depicted in FIG. **17**, the tip **456** of the jar slide **451** contacts the housing slide **464** such that both the jar slide and the housing slide are respectively urged against the biasing force of the jar slide biasing member **458** and the housing slide biasing member **468** toward respective open positions in which liquid condensate is allowed to pass through the corresponding condensate passageway. A channel

defined by adjacent vanes **455** of the valve stem allows the condensate to pass through the condensate passageway when the valve stem is inserted into the condensate passageway. Thus, when the collection jar **440** is attached to the filter housing **420**, each valve assembly **450a,b** automatically assumes the open position to allow condensate to pass from the filter housing through the corresponding condensate passageway and into the collection jar.

[0122] When the collection jar **440** is detached from the filter housing **420**, as depicted in FIG. **18**, the tip **456** of the jar slide **451** no longer contacts the housing slide **464**, such that both the jar slide and the housing slide are respectively urged by the biasing force of the jar slide biasing member **458** and the housing slide biasing member **468** toward the closed position in which the corresponding condensate passage is blocked, thus preventing liquid condensate from passing therethrough. As described above, in the closed position, the flange **452** of the jar slide sealingly abuts an interior surface of a portion of the collection jar cover **441** surrounding the corresponding condensate passageway to block it, and thus preventing condensate from spilling out of the collection jar through the condensate passageway. The elastomeric jar seal **459** may be disposed between the flange **452** and the interior surface of the collection jar to ensure a fluid tight seal when the valve assembly is in the closed position.

[0123] Similarly, the housing slide **464** sealingly abuts an interior surface of a portion of the filter housing base **421** surrounding the corresponding condensate passageway to block it, and thus preventing condensate from leaking out of the filter housing through the condensate passageway. The elastomeric housing seal **469** may be disposed between the housing slide **464** and the interior surface of the filter housing base to ensure a fluid tight seal. Thus, when the collection jar **440** is detached from the filter housing **420**, each valve assembly **450a,b** automatically moves to the closed position, thus obstructing the corresponding condensate passageway, and therefore preventing condensate from leaking out of both the filter housing and the collection jar through the corresponding condensate passageway.

[0124] In each of the above implementations of the respiratory filter and condensate management apparatus, a user is able to detach the collection jar from the filter housing when the user wants to empty the condensate collected within the reservoirs. Upon such detachment, the self-sealing valve assemblies automatically close to prevent accidental spillage of condensate.

[0125] While the respiratory filter and condensate management apparatus has been described in terms of what may be considered to be specific aspects, the present disclosure is not limited to the disclosed aspects. Additional modifications and improvements to the respiratory filter and condensate management apparatus may be apparent to those skilled in the art. Moreover, the many features and advantages of the disclosure are apparent from the detailed specification, and thus, it is intended by the appended claims to cover all such features and advantages of the disclosure which fall within the spirit and scope of the disclosure.

[0126] Further, it is not desired to limit the disclosure to the exact construction and operation illustrated and described, and accordingly, all suitable modifications and equivalents may be resorted to, falling within the scope of the disclosure. The present disclosure should therefore be considered as illustrative and not restrictive. As such, this disclosure is intended to cover various modifications and similar arrangements included within the spirit and scope of the claims, which should be accorded their broadest interpretation so as to encompass all such modifications and similar structures.

Claims

1. A respiratory filter and condensate management apparatus for a breathing circuit, the apparatus comprising: a filter housing including a patient side, a ventilator side, and a base extending from the patient side to the ventilator side, the patient side having an air inlet port configured to receive a flow of expiratory air from a patient, the ventilator side having an air outlet port configured to

output the flow of expiratory air to a ventilator, and the base including at least one base drain; a filter provided within the filter housing and located in a flow path of the expiratory air between the air inlet port and the air outlet port; a collection jar attached to the filter housing, the collection jar including a cover and a liquid reservoir, the cover including at least one cover drain, the liquid reservoir configured to receive liquid formed by condensation in the flow of expiratory air within the filter housing, and the collection jar configured to be detached from the filter housing; at least one liquid passageway formed by the at least one base drain and the at least one cover drain; and at least one valve assembly configured to move to a closed position when the collection jar is detached from the filter housing, wherein passage of liquid from the filter housing through the base drain and passage of liquid from the collection jar through the cover drain is prevented when the at least one valve assembly is in the closed position.

2. The respiratory filter and condensate management apparatus according to claim 1, wherein the at least one base drain includes a patient side base drain and a ventilator side base drain.

3. The respiratory filter and condensate management apparatus according to claim 2, wherein the at least one cover drain includes a patient side cover drain and a ventilator side cover drain.

4. The respiratory filter and condensate management apparatus according to claim 3, wherein the at least one liquid passageway includes a patient side liquid passageway and a ventilator side liquid passageway, the patient side liquid passageway formed by the patient side base drain and the patient side cover drain, and the ventilator side liquid passageway formed by the ventilator side base drain and the ventilator side cover drain.

5. The respiratory filter and condensate management apparatus according to claim 4, wherein the at least one valve assembly includes a patient side valve assembly and a ventilator side valve assembly.

6. The respiratory filter and condensate management apparatus according to claim 1, wherein the at least one valve assembly is configured to automatically move to an open position when the filter housing is attached to the collection jar, and wherein the at least one valve assembly is configured to automatically move to the closed position when the filter housing is detached from the collection jar.

7. The respiratory filter and condensate management apparatus according to claim 1, wherein the at least one valve assembly comprises a housing slide configured to close the base drain of the filter housing when the collection jar is detached from the filter housing.

8. The respiratory filter and condensate management apparatus according to claim 7, wherein the at least one valve assembly further comprises a housing slide biasing member configured to bias the housing slide toward the closed position.

9. The respiratory filter and condensate management apparatus according to claim 8, wherein the housing slide biasing member is a compression spring.

10. The respiratory filter and condensate management apparatus according to claim 8, wherein the at least one valve assembly further comprises an elastomeric housing seal configured to provide a liquid-tight seal between the housing slide and the base drain when the at least one valve assembly is in the closed position.

11. The respiratory filter and condensate management apparatus according to claim 1, wherein the at least one valve assembly comprises a jar slide configured to close the cover drain of the collection jar when the collection jar is detached from the filter housing.

12. The respiratory filter and condensate management apparatus according to claim 11, wherein the at least one valve assembly further comprises a jar slide biasing member configured to bias the jar slide toward the closed position.

13. The respiratory filter and condensate management apparatus according to claim 12, wherein the jar slide biasing member is a compression spring.

14. The respiratory filter and condensate management apparatus according to claim 12, wherein the at least one valve assembly further comprises an elastomeric jar seal configured to provide a liquid-

tight seal between the jar slide and the cover drain when the at least one valve assembly is in the closed position.

15. The respiratory filter and condensate management apparatus according to claim 1, wherein the at least one valve assembly comprises a housing slide and a jar slide, the housing slide configured to close the base drain of the filter housing when the collection jar is detached from the filter housing, and the jar slide configured to close the cover drain of the collection jar when the collection jar is detached from the filter housing.

16. The respiratory filter and condensate management apparatus according to claim 15, wherein the jar slide further comprises a stem configured to abut the housing slide when the at least one valve assembly is in an open position.

17. The respiratory filter and condensate management apparatus according to claim 1, further comprising an impact pad provided within the filter housing, the impact pad operable to remove liquid from the flow of expiratory air.

18. The respiratory filter and condensate management apparatus according to claim 17, wherein the impact pad is an electrostatic pad.

19. The respiratory filter and condensate management apparatus according to claim 1, wherein the filter is pleated.

20. The respiratory filter and condensate management apparatus according to claim 1, further comprising a latch mechanism configured to releasably attach the filter housing to the collection jar, wherein the latch mechanism is operable to permit detachment of the collection jar from the filter housing without disconnecting the filter housing from the breathing circuit.
